Research Plan

Conformal-Controlled Adaptive Filtering: distribution-free guarantees and coverage-driven step-size control, benchmarked on Gaussian noise and beyond

Full pipeline: novelty positioning, mathematical modelling, theory, baselines, experiments, statistics, venue strategy, timeline. Prepared 18 October 2026.

One-paragraph summary. Classical adaptive filters (LMS, NLMS, RLS, Kalman) are provably well-behaved under Gaussian noise, but they return a point estimate only, and their step size or forgetting factor is tuned by hand or by heuristics that are already heavily explored. This plan proposes **CCAF**: an adaptive filter that (1) outputs prediction intervals with a **distribution-free, long-run coverage guarantee** via online conformal inference, and (2) **closes the loop**, using the conformal miscoverage state as the control signal for the step size. Gaussian noise is the nominal benchmark, where the filter must match the oracle Kalman/Wiener bound; heavy tails, variance jumps and abrupt plant changes are where it should win.

Honest limits you should know before starting.

- **Nobody can guarantee a completely novel idea or acceptance.** What this plan does is pick a direction where my searches (18 Oct 2026) found no direct prior art, and give you a protocol (Section 9) to confirm or kill that within two weeks.
- **ACM/IEEE TON is the wrong venue.** It is a computer-networking journal. The right top venues are IEEE Trans. Signal Processing and peers (Section 11).
- **A new variable-step-size LMS alone is not publishable at the top level.** That literature is saturated (sigmoid, error-feedback, nonparametric, gradient-adaptive, convex-combination and many more variants). The novelty here is the conformal coupling plus theory, not another step-size formula.

1. Landscape and the gap

What exists, and what I could not find. Items marked *verify* are from memory or partial search and must be confirmed in your own literature review.

Area	Representative prior art	What it does not do
LMS / NLMS / RLS / APA theory	Haykin; Sayed (*Adaptive Filters*, 2008); Diniz. Mean and mean-square analysis under the independence assumption.	Gives MSE statements only. No finite-sample, distribution-free uncertainty statement about the output.
Variable step-size (VSS) filters	Kwong and Johnston (1992); sigmoid, error-feedback and nonparametric VSS-NLMS (Benesty et al., 2006); gradient-adaptive step sizes.	Step size driven by error power, SNR estimates or gradients. None of the ones I found is driven by a calibrated coverage signal.
Combination of filters	Convex and affine combinations of a fast and a slow filter (Arenas-Garcia et al.; Sayed and co-authors).	Doubles the filter cost, and has no interval or coverage output.
Online conformal inference	Adaptive Conformal Inference, ACI (Gibbs and Candes, 2021) and many 2023 to 2026 extensions (localized, multi-scale, semi-bandit).	Developed for ML predictors. In my searches I found no work applying it to LMS-family weight tracking.

Area	Representative prior art	What it does not do
Conformal + state estimation	*Conformalized Kalman Filters for State Estimation with Trustworthy Confidence Regions* (arXiv 2609.27506, found in search; *verify details*).	Targets Kalman-type state estimation. Does not address LMS/RLS weight tracking or use coverage to drive the step size (as far as I could see; you must read it closely).
Learned adaptive filters	Meta-learned optimizers and deep-learning hybrids for adaptive filtering (*verify current state*).	Need training data and lose the clean Gaussian-case theory. A classical, provable method remains attractive.

2. The proposed idea: CCAF

Setting. Unknown time-varying linear system observed through noise (system identification / noise cancellation):

$$d(n) = w_o(n)^T u(n) + v(n), \quad w_o(n+1) = w_o(n) + q(n), \quad v(n) \sim N(0, \sigma_v^2), \; q(n) \sim N(0, \sigma_q^2 I)$$

Gaussian noise is the **nominal model**: there the Kalman filter with known (σ_v^2, σ_q^2) is the MMSE-optimal tracker and gives an oracle lower bound. Real data violate the model (heavy tails, outliers, variance jumps), which is where distribution-free guarantees matter.

2.1 Algorithm sketch (one iteration)

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1. Predict: y^(n) = w(n)' u(n); interval y^(n) +/- q^(n) where q^(n) is the (1 - a_n) empirical quantile
of recent normalized scores times the running scale s^(n)
2. Observe d(n); a-priori error e(n) = d(n) - y^(n); score r(n) = |e(n)| / s^(n)
3. Miss flag: m(n) = 1 if r(n) > q-level, else 0
4. ACI update: a_{n+1} = a_n + gamma * (alpha - m(n)) [Gibbs-Candes recursion]
5. Control: c(n) = clip((alpha - a_n)/alpha, 0, 1); mu(n) = mu_min + (mu_max - mu_min) * c(n)^p
6. Filter: w(n+1) = w(n) + mu(n) * e(n) * u(n) / (eps + ||u(n)||^2) [NLMS, 0 < mu_max < 2]
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The control function in step 5 is a design choice to be optimised and ablated (monotone, bounded, Lipschitz, mu_max below the NLMS stability limit). The intuition: after an abrupt change, misses accumulate, the ACI state a_n drops, the step size rises and the filter re-converges quickly; in steady state the step size relaxes to mu_min.

2.2 Target contributions (your paper's claims)

ID	Claim	Difficulty	Note
C1	**Coverage for any coupled filter.** The long-run miscoverage error is bounded by $O(1/(\gamma T))$ for arbitrary score sequences, even though the filter depends on the coverage state.	Easy to moderate	Follows from the ACI guarantee, which makes no distributional assumption. Prove carefully that the coupling does not break the preconditions (bounded scores, clipping of a_n).
C2	**Nominal efficiency.** Under stationary Gaussian noise, CCAF's steady-state EMSE is within a small factor of NLMS with mu_min (and close to the oracle), with the factor controlled by the false-alarm rate.	Hard	Needs a switching-step-size analysis. Likely under standard simplifications (independence assumption, small mu). State them openly.
C3	**Tracking.** After an abrupt change of size Δ, the detection delay is bounded in terms of γ, α and Δ/σ_v, and the MSD recovers at a stated rate.	Hard	ACI drift after consecutive misses is explicit: a_n falls by $\gamma(1-\alpha)$ per miss. This gives a clean delay expression.
C4	**Robustness.** Under Student-t and outlier-contaminated noise, Gaussian intervals (RLS/Kalman covariance) undercover, while CCAF holds nominal coverage with intervals only modestly wider.	Empirical	Report coverage and interval score, not only MSE.

Why this can be strong. It joins two mature fields, delivers something classical filters cannot (a guaranteed interval), keeps the Gaussian-case theory as the anchor, and the loop-closing step is genuinely different from error-power-driven VSS rules. **What can sink it:** reviewers may call it "ACI plus NLMS". The defence is C2 and C3: real theorems about the coupled system, plus a clear ablation showing the feedback matters (coupled vs decoupled).

3. Mathematical modelling to write out in the paper

3.1 Baseline derivations (full, not just cited)

- Wiener solution: $w^* = R^{-1} p$, with $J_{min} = \sigma_d^2 - p^T R^{-1} p$, $R = E[u\, u^T]$, $p = E[d\, u]$.
- LMS: $w(n+1) = w(n) + \mu\, e(n)\, u(n)$. Mean stability: $0 < \mu < 2/\lambda_{max}$. Mean-square stability (independence assumption): $\Sigma_i \mu\lambda_i / (2 - \mu\lambda_i) < 1$. For small μ: excess MSE $\approx \mu\, J_{min}\, tr(R)/2$, misadjustment $M \approx \mu\, tr(R)/2$.
- NLMS: $\mu/(\varepsilon + ||u||^2)$ normalisation; stable for $0 < \mu < 2$. RLS via the matrix inversion lemma with forgetting factor λ; excess MSE roughly proportional to $(1-\lambda)M/(1+\lambda)$ (check the exact form in Haykin before quoting).
- Tracking trade-off: under the random-walk model, tracking MSD has a noise term growing with μ and a lag term growing like σ_q^2/μ, so an optimal fixed μ exists; a good variable step size approaches it without knowing σ_q^2.
- Kalman filter on the weight state: the MMSE benchmark under Gaussian noise. Use it as the oracle line in every plot.

3.2 Conformal layer

- Score: $r(n) = |e(n)|/\hat{s}(n)$ with $\hat{s}(n)$ a running scale estimate (exponentially weighted RMS of past errors).
- ACI recursion: $a_{n+1} = a_n + \gamma(\alpha - m(n))$. Guarantee (Gibbs and Candes): $|(1/T)\, \Sigma\, m(n) - \alpha| \leq (\max(a_1, 1-a_1) + \gamma)/(\gamma T)$ for arbitrary sequences. Confirm the exact statement and conditions from the paper before citing.
- Interval for the observation $d(n)$. If you want intervals for the clean output or for the weights, that is a harder, separate extension; list it as future work.
- Interval score (Winkler): width + $(2/\alpha)$(distance outside interval) as a single sharpness-and-coverage metric.

4. Baseline comparison set

Choose baselines that span cost and performance and include oracles. Tune every baseline fairly on a separate validation seed set; reviewers check this.

Group	Algorithms	Why included
Oracles (not realisable)	Wiener (known R, p); Kalman with known σ_v^2, σ_q^2	Lower bound on achievable MSD/EMSE under Gaussian noise
Classical fixed-parameter	LMS (slow and fast μ), NLMS, Sign-error LMS, RLS (λ = 0.999, 0.99), Affine Projection (K = 2, 4)	Standard references every reviewer expects
Variable step-size	Kwong-Johnston VSS-LMS; nonparametric VSS-NLMS; sigmoid/error-feedback VSS; gradient-adaptive step size	Direct competitors for the step-size idea
Combination	Convex combination of fast + slow NLMS	Strong tracking baseline, but at double cost
Noise-adaptive Kalman	Sage-Husa adaptive Kalman (estimates noise covariances online)	Fair comparison when σ_v^2 is unknown
Uncertainty baselines	Gaussian interval from RLS/Kalman covariance; static split-conformal on NLMS; ACI on fixed-μ NLMS (decoupled ablation); Conformalized Kalman Filter if reproducible	Isolates what coupling adds
Recent (add after your review)	2024 to 2026 adaptive-filter papers found in your systematic search	Top journals require current baselines

5. Experimental design

ID	Scenario	Key parameters (starting values)	Tests claim
S1	Stationary Gaussian, white input	M = 32 taps; SNR in {-5, 0, 5, 10, 15, 20, 30} dB; T = 20,000	C2, nominal efficiency
S2	Stationary Gaussian, coloured input	AR(1) input with ρ in {0.5, 0.9, 0.99} (large eigenvalue spread)	C2
S3	Abrupt plant change	Weights sign-flipped / rescaled at n = 8,000; Δ sweep	C3 detection delay
S4	Random-walk plant	σ_q^2 sweep over three decades	Tracking optimality vs Kalman
S5	Time-varying noise variance	Piecewise σ_v^2 jumps of 5x to 20x	C1, C3
S6	Non-Gaussian stress	Student-t (ν = 2.5, 3, 5); Gaussian-mixture outliers (5 to 10%, scale 10x); impulsive noise	C4
S7	Real data (at least two)	EEG: EEGdenoiseNet semi-synthetic artifacts; ECG: MIT-BIH with the Noise Stress Test Database; optionally an acoustic echo path	Real-world validation

- **Monte Carlo:** 300 independent runs per cell; the same random seeds across all algorithms so comparisons are paired.
- **Metrics:** MSD learning curves (dB), steady-state EMSE, time to reach within 1 dB of floor after a change, empirical coverage (overall and in sliding windows), mean interval width, Winkler interval score, flops per iteration and runtime.
- **Ablations:** γ and α sensitivity; score normalisation; control function g and exponent p; coupled vs decoupled (the key ablation); filter length mismatch; mu_max.
- **Real-data protocol:** use public, citable datasets with a known clean reference so MSE is measurable; fix preprocessing and publish it.

6. Statistical analysis protocol

Question	Method	Report
Is CCAF better than baseline X in one scenario?	Paired Wilcoxon signed-rank on per-run steady-state EMSE (dB), same seeds	Median difference, p-value, Cliff's delta or Cohen's d
Many comparisons at once	Holm-Bonferroni correction across all pairwise tests	Adjusted p-values
Ranking across scenarios	Friedman test with Nemenyi post-hoc, using scenario x SNR cells as blocks (Demsar, 2006)	Critical-difference diagram
Uncertainty of a mean curve	BCa bootstrap, 10,000 resamples	95% bands on every learning curve
Is coverage truly nominal?	Clopper-Pearson interval on empirical coverage per run set; sliding-window coverage curves	Coverage vs 1-α with exact CIs
Practical vs statistical significance	With 300 runs, p-values become tiny for trivial gaps	Pre-set a practical threshold (for example 0.5 dB) and report effect sizes first

Pre-register this analysis plan (for example on OSF) before running the final experiments, and never tune hyper-parameters on the final test seeds.

7. Theory workplan

- **Step 1 (week 1 to 2 of theory):** prove C1 formally. Verify ACI preconditions under feedback coupling. This gives you a safe, publishable core.
- **Step 2:** analyse the coupled system under the independence assumption and small-step approximations; derive steady-state EMSE with a bound on the step-size fluctuation caused by false alarms (C2).
- **Step 3:** derive the detection-delay and recovery bound after an abrupt change (C3). Use the explicit ACI drift.

- **Step 4:** validate every closed-form result against simulation; plot theory vs Monte Carlo on the same axes. Disagreement beyond a few tenths of a dB signals an error or a hidden assumption.
- **If C2/C3 prove too hard:** narrow to C1 + C4 + a rigorous simplified analysis (for example a two-state step-size switch), and aim for the shorter-format venue first.

8. Reproducibility and integrity

- Public repository (Python/NumPy; fixed seeds, one config file per scenario) archived with a DOI (Zenodo). Provide a one-command script that regenerates every figure.
- Report all failures honestly, including scenarios where CCAF is worse or only matches the baselines. Matching the oracle in S1 is a success, not a failure.
- Every result in the paper must come from code you ran. Do not copy numbers from other papers into your tables unless you cite and label them as such.
- Use plagiarism checks and disclose any AI assistance according to the target journal's policy.

9. Novelty verification protocol (do this in weeks 1 to 2)

Goal: either confirm the gap or discover the closest prior art early. Document every search in a spreadsheet (date, database, query string, hits, decision).

Step	Action
Databases	IEEE Xplore, Scopus or Web of Science, arXiv, Google Scholar, Semantic Scholar
Query strings	(conformal OR "conformal prediction") AND (LMS OR NLMS OR RLS OR "adaptive filter"); "adaptive conformal inference" AND ("step size" OR "adaptive filtering"); coverage AND "variable step-size"; conformal AND ("system identification" OR "echo cancellation" OR "noise cancellation"); "uncertainty quantification" AND "adaptive filtering"
Citation chasing	Forward-citations of Gibbs and Candes (2021) and of the Conformalized Kalman Filter paper; backward references of any close hit
Read closely	The Conformalized Kalman Filter paper end to end: list exactly what it does and does not cover; write a one-page comparison
Decision rule	If a paper already has coverage-guaranteed LMS-family intervals **and** coverage-driven step-size control, pivot to Plan B or C (Section 12). If only one of the two is covered, differentiate on the other plus theory C2/C3.
Advisor check	Show the one-page gap statement to a faculty member or researcher in signal processing before investing months. This is the single best way to catch what searches miss.

10. Timeline (24 weeks)

Weeks	Phase	Deliverable	Exit gate
1 to 2	Novelty check and scoping	Search log, one-page gap statement, closest-paper comparison	Gap confirmed or pivot decided
3 to 5	Literature review and baselines	Annotated bibliography (40 to 60 papers); baselines implemented and unit-tested	Baselines reproduce textbook curves
6 to 9	CCAF design and theory C1	Working algorithm; proof of C1; first simulations in S1, S3	Coupled beats decoupled in S3
10 to 13	Theory C2 and C3	Derivations with simulation agreement	Theory within a few tenths of a dB
14 to 17	Full experiments S1 to S6	All scenarios, 300 runs, tuned baselines	Results frozen, analysis pre-registered
18 to 19	Real data S7 and ablations	EEG, ECG results; ablation table	At least two real datasets done
20 to 22	Writing	Full draft, figures, complexity table, appendix proofs	Draft complete
23	Internal review	Faculty or peer review; fix gaps; preprint on arXiv	Reviewer-style critique addressed
24	Submission	Cover letter, reproducibility package	Submitted

11. Paper structure and venue strategy

Section	Content	Share
Introduction	Problem, gap, four contributions (C1 to C4), paper roadmap	8%
Related work	By family (Section 1 table), ending with what is missing	7%
Model and algorithm	System model, CCAF, design rationale	15%
Theory	C1 to C3 with assumptions listed; proofs in appendix	25%
Experiments	Setup, S1 to S7, ablations, complexity	30%
Discussion and limits	Where it fails, assumptions, future work	7%
Conclusion and appendix	Summary; full proofs; extra plots	8%

Tier	Venue	Fit and note
Primary target	IEEE Transactions on Signal Processing	Best fit for theory plus algorithm. High bar; needs C2/C3 with solid analysis.
Strong alternatives	Signal Processing (Elsevier); Digital Signal Processing; IEEE Trans. on Circuits and Systems I/II (if you add a low-complexity hardware angle)	Slightly lower bar; still respected.
Fast first step	IEEE Signal Processing Letters (about 5 pages) or a signal-processing conference (ICASSP, EUSIPCO)	Publish the core (C1 + C4) quickly to establish priority, then extend to a full journal paper. Check current deadlines.
Domain angle	IEEE Trans. on Biomedical Engineering or Biomedical Signal Processing and Control	If the EEG/ECG results become the headline.
Not a fit	IEEE/ACM Transactions on Networking	Networking journal; this topic is out of scope.

Reviewer-risk checklist: novelty clearly separated from ACI and from the Conformalized Kalman Filter; assumptions explicit; theory matches simulation; baselines current and fairly tuned; real data; code available; limitations stated.

12. Risks and fallback plans

Risk	Likelihood / impact	Mitigation
Scooped or prior art found	Medium / high	Weekly arXiv alerts on key terms; the early novelty check; pivot options below
Theory C2/C3 too hard	Medium / medium	Narrow to C1 + C4 + simplified analysis; target Signal Processing Letters first
Gains small under Gaussian noise	High / low	Expected. Frame as: matches the oracle when Gaussian, wins under departures
Reviewer says incremental	Medium / high	Decoupled-vs-coupled ablation plus C2/C3 theorems; show what feedback uniquely enables
Compute or time overrun	Low / low	Fix T and run counts early; parallelise Monte Carlo

Fallback ideas (unverified; run the same novelty check before committing):

- **Plan B:** anytime-valid sequential change detection (e-process or CUSUM with exact Gaussian likelihood ratios) controlling a step-size reset with a provable false-alarm bound.
- **Plan C:** low-complexity, fixed-point/FPGA-friendly variant of the coupled filter with a quantisation-error analysis (suits the circuits venues).

13. First 14 days: concrete actions

- Day 1 to 3: read Gibbs and Candes (ACI) and the Conformalized Kalman Filter paper; read the LMS/NLMS/RLS chapters you need in Sayed or Haykin.
- Day 4 to 7: run the protocol in Section 9 and fill the search log.
- Day 8 to 10: implement NLMS, RLS, Kalman oracle and ACI in Python; reproduce one textbook learning curve and one ACI coverage plot.
- Day 11 to 12: implement the CCAF loop; run S3 (abrupt change) with coupled vs decoupled.
- Day 13 to 14: write the one-page gap statement and the first-results note; send both to a faculty member for feedback. Decide go or pivot.

References to start from (verify details before citing)

[1] Gibbs, I. and Candes, E. Adaptive conformal inference under distribution shift. NeurIPS, 2021.

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